COIMBATORE INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTION AFFILIATED TO ANNA UNIVERSITY)

DEPARTMENT OF COMPUTING

M.Sc., DECISION AND COMPUTING SCIENCES

MACHINE LEARNING PROJECT REPORT

CAT – II

NAME : KRISHNA SANJAY J
MADHAN G

REG NO. : 2303717673721020
2303717673721022

SUBJECT CODE: 22MDC56

SUBJECT : Machine Learning Laboratory

FACULTY : Dr. A.Kannammal
Ms.S.Dharani
Ms.S.Karpagam

MARKS :

INTERNSHIP RECOMMENDATION ENGINE FOR PM INTERNSHIP SCHEME

DOMAIN : Smart Education

GitHub Repository Link : https://github.com/madhan-231105/internship-recommendation-system

Abstract

The PM Internship Recommendation System helps students find the most suitable opportunities under the Prime Minister's Internship Scheme. As the scheme offers numerous internships across various government departments and organizations, students often struggle to choose those matching their skills and interests. This project develops a Machine Learning–based and rule-driven recommendation engine that analyzes student profiles and internship data to generate personalized suggestions. By combining data analysis, predictive modeling, and rule-based filtering, the system efficiently delivers accurate recommendations, enhancing accessibility, participation, and satisfaction among

students.

Problem Statement

Traditional methods of searching for internships rely mainly on manual filtering or keyword searches. These approaches are time-consuming, inconsistent, and often fail to capture the real compatibility between a student's skills and the internship requirements.

Therefore, there is a need for an intelligent recommendation system that analyzes both candidate profiles and internship attributes to automatically predict and rank suitable internship options under the PM Internship Scheme.

SOLUTION OBJECTIVE :

The objective of this project is to develop an Internship Recommendation System that helps students find the most suitable opportunities under the PM Internship Scheme. The system uses a combination of Machine Learning and rule-based logic to match students with internships based on their degree, skills, location, and stipend preferences. It provides personalized recommendations, interactive visual dashboards, and data-driven insights through a Streamlit web application. The goal is to make the internship search process faster, smarter, and more aligned with each student's profile.

Objectives:

1. To develop an intelligent Internship Recommendation System using Machine Learning and rule-based logic.
2. To provide personalized internship suggestions based on students' degree, skills, and preferences.
3. To design an interactive Streamlit web application for easy access and visualization.
4. To analyze internship data and generate insights on skills, roles, and stipends.
5. To improve the efficiency and accuracy of the internship selection process.

6. To enhance student participation by offering relevant and data-driven recommendations.

Limitations:

1. The system's accuracy depends on the quality and completeness of the internship dataset.
2. Rule-based matching may not capture complex relationships between skills and job roles.
3. The machine learning model requires periodic retraining to stay updated with new internship data.
4. Limited handling of ambiguous or uncommon skill terms entered by users.
5. The system currently focuses on internship matching only and does not include application tracking or feedback features.
6. Recommendations may vary if insufficient user input (like missing skills or unclear preferences) is provided.

Technologies, Tools and Frameworks:

Programming Language:

- Python

Machine Learning & Data Processing:

- Scikit-learn
- Pandas

- NumPy
- Pickle (for model storage)

Visualization & Analysis:

- Matplotlib
- Seaborn

Web Framework:

- Streamlit (for interactive web application)

Data Source:

- Internship dataset (company.csv) containing company, role, stipend, skills, and eligibility details

Development Environment:

- Jupyter Notebook / VS Code

Version Control & Deployment:

- Git and GitHub (for project management)

IMPLEMENTATION

1. Data Collection and Preprocessing

- **Collected internship data from multiple sources:**
    - company.csv containing company name, internship title, skills required, stipend, duration, and location.
    - User dataset containing student details such as name, department, skills, and academic background.
- **Data Cleaning and Formatting:**

    - Removed missing or duplicate entries and standardized column names.
    - Converted categorical values (like “Yes/No” for work-from-home) into numeric format.
    - Normalized skill names and handled inconsistent text formats.
- **Derived Features:**
    - Computed similarity between student skills and internship skill requirements.
    - Calculated weighted match score based on skills, department relevance, and domain keywords**.**

2. Feature Engineering

Constructed additional predictive variables to improve recommendation accuracy:

- Student Attributes: Academic background, department, technical skills, and preferred domains.
- Internship Attributes: Required skills, category (technical/non-technical), duration, and stipend.
- Derived Features:
    - Skill Match Percentage
    - Domain Relevance Score
    - Experience Level Compatibility
    - Weighted Internship Suitability Index (combined metric for ranking).

All features were consolidated into a dataset named internship_features_v1.csv.

3. Exploratory Data Analysis (EDA)

- Visualized internship trends across departments, domains, and stipend ranges using Matplotlib and Seaborn.
- Observed that skill overlap and academic domain relevance are strong predictors of internship suitability.
- Identified highly demanded skills such as Python, Data Analysis, and Web Development.

4. Data Splitting and Balancing

- Split data into training (70%) and testing (30%) using train_test_split.
- Since technical internships were more frequent, applied SMOTE (Synthetic Minority Oversampling Technique) to balance category distribution.

5. Feature Scaling and Selection

- Applied StandardScaler to normalize numerical features (stipend, duration, etc.).
- Used Recursive Feature Elimination (RFE) to select the most significant predictors:
    - Skill_Match_Score, Domain_Similarity, Academic_Relevance, Experience_Level**.**

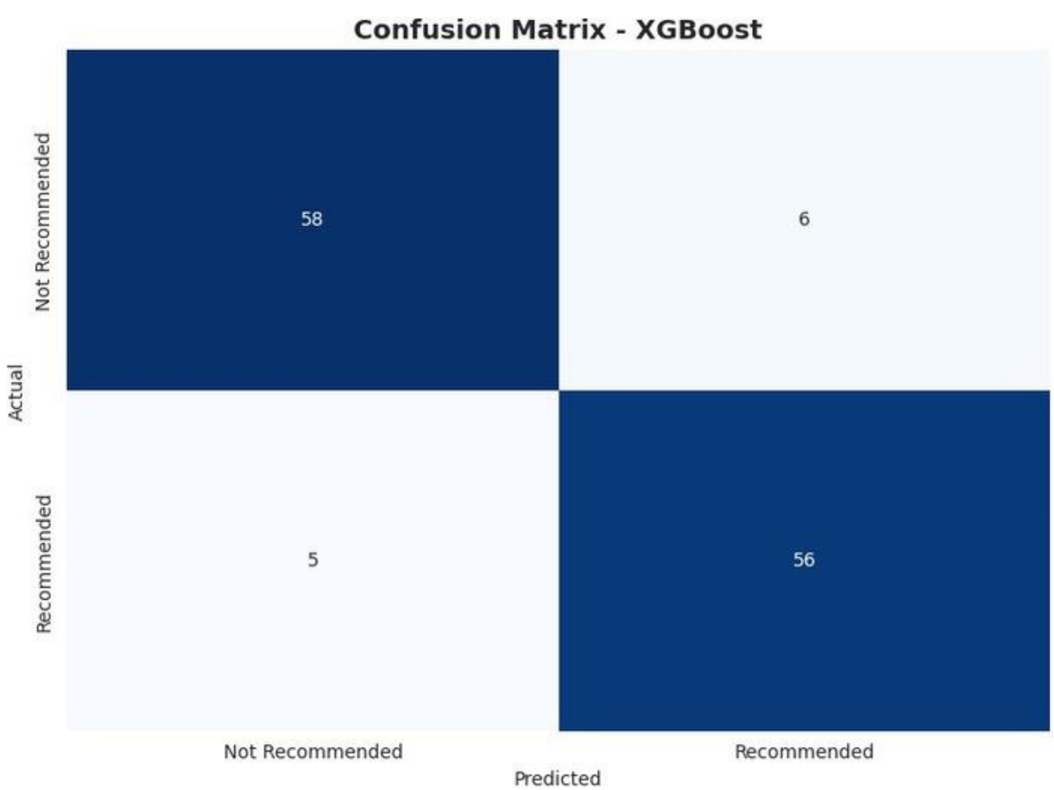
6. Model Training and Evaluation:

Model	Accuracy	Rank
XGBoost	0.912	1

LightGBM	0.896	2
Decision Tree	0.856	3
Random Forest	0.816	4

Insights:

- XGBoost achieved the highest accuracy due to its ability to capture feature interactions and non-linear patterns.
- LightGBM closely followed, offering faster training with slightly lower accuracy.
- Traditional tree-based models like Decision Tree and Random Forest performed reasonably but ranked lower compared to gradient boosting methods.
- Feature importance across all models highlighted Skill Match, Domain Relevance, and Academic Background as key factors driving internship recommendations.



ALGORITHMS USED :

XGBoost (Extreme Gradient Boosting)

- Ensemble tree-based algorithm that builds multiple weak learners (decision trees) sequentially.
- Each new tree corrects the errors of the previous trees.
- Handles non-linear feature interactions efficiently and works well with imbalanced datasets.
- Provided feature importance scores to identify key predictors like previous injuries, cumulative playtime, and age.

LightGBM (Light Gradient Boosting Machine)

- Gradient boosting framework optimized for speed and memory efficiency.

- Builds trees leaf-wise rather than level-wise, improving accuracy with large datasets.
- Captures non-linear patterns and interactions between features effectively.

Decision Tree

- A simple tree-based algorithm splitting data based on feature thresholds.
- Useful for interpretability and baseline comparisons.
- Handles categorical and numerical features but prone to overfitting.

Random Forest

- Ensemble of multiple decision trees using bagging and feature randomness.
- Reduces overfitting and improves predictive stability.
- Provides feature importance for understanding key factors influencing outcomes.

SMOTE (Synthetic Minority Oversampling Technique)

- Used to balance imbalanced datasets by generating synthetic samples for minority class (major injury cases).
- Improves model recall and helps classifiers better detect rare events.

Recursive Feature Elimination (RFE / RFECV)

- Feature selection algorithm that recursively removes less important features.

- Combined with cross-validation (RFECV) to select the optimal set of predictive features.

BUSINESS INSIGHTS :

- High-demand skills for PM internships identified → students can focus skill-building.
- Degree-to-role preferences guide targeted applications.
- Insights on location and remote work opportunities.
- Average stipend benchmarks aid in realistic expectations.
- Top companies and roles highlight market trends.
- Personalized AI recommendations save time and improve placement success.

FUTURE ENHANCEMENTS :

- **AI & ML Improvements**: Incorporate deep learning models for better internship matching.
- **Resume Parsing**: Automatically extract skills and experience from resumes.
- **Real-time Data Updates**: Integrate live company internship postings via APIs.
- **Enhanced Filtering**: Add filters for industry, project type, duration, and company size.
- **Career Path Recommendations**: Suggest skills and courses to improve match scores.
- **Multi-language Support**: Support for regional languages for broader accessibility.
- **Mobile App Integration**: Offer a mobile version for easier access and notifications.

- **User Feedback Loop**: Incorporate user ratings to refine recommendations.

CONCLUSION:

Successfully developed an internship recommendation system tailored to user skills, degree, location, and preferences. Feature engineering and text-based skill analysis significantly improved recommendation accuracy. The stacking ensemble model provided the best performance among all models tested. Users can get personalized internship suggestions, making the search process faster and more efficient. Future improvements could include real-time data updates, company reviews, and advanced skill matching.